

VAR1 NET POWER (LASSO) CV R² = 0.9698 CV MSE: 0.3450 • Deg 5 (195 active)	VAR2 THERMAL ANOMALY (RIDGE) CV R² = 0.9942 CV MSE: 0.2685 • Deg 9 (norm 20.55)	BASELINE → FINAL GAIN +0.817 / +0.913 R² 96.4% & 99.4% MSE reduction	CONSTRAINT COMPLIANCE Strict Polynomial 5-Fold CV • 0 Data Leakage
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1. Introduction & Overview

This report documents the polynomial regression models built for two distinct engineering prediction problems assigned to my roll number (BT2024043):

- Problem 1 (var1) — Steam Turbine Net Power Score:** Predicting power output from 6 operational parameters (steam valve, coolant flow rate, pump pressure, blade pitch, exhaust rate, and inlet pressure).
- Problem 2 (var2) — Subterranean Thermal Anomaly Score:** Mapping subsurface temperatures across a 3D geological reservoir from spatial offsets (x_1 : East-West, x_2 : North-South, x_3 : Depth).

The assignment required using **strictly polynomial regression** without non-polynomial architectures. When I initially tested the starting hints mentioned in the prompt (degree 3 on x_1 – x_3 for var1, and degree 4 on x_1 for var2), the validation score was poor ($R^2 \approx 0.15$ and 0.08). To find the true models, I followed a four-stage progression:

Four-Stage Trajectory: **M1 (Baseline):** Tested PDF hints literally with OLS → heavy underfitting ($R^2 \approx 0.08 - 0.15$). • **M2 (OLS Sweep):** Added all features across degrees → massive jump ($R^2 \approx 0.92 - 0.99$), but hit unregularized variance limit. • **M3+ (Regularized — Final):** Applied Lasso (L_1) to prune 57.8% of turbine terms, and Ridge (L_2) to smoothly stabilize 3D heat fields → peak generalization (var1 CV $R^2 = 0.9698$, MSE = 0.3450; var2 CV $R^2 = 0.9942$, MSE = 0.2685). • **M4 (Overfit Demo):** Pushed degrees to 8 & 15 with unregularized OLS → training error near zero, but validation MSE exploded to 1.57×10^{11} .

2. Exploring the Datasets

2.1 Turbine Net Power Score (var1): Power generation follows thermodynamic laws (Brayton cycle). Net work depends on 6 operational controls (x_1 – x_6) normalized in $[-1.0, 1.0]$. The dataset contains 1,000 training samples with target mean $\mu = 0.76$, std $\sigma = 3.31$. Inputs interact, but arbitrary 5-way cross products rarely correspond to physical processes. This pointed to a **sparse polynomial**.

2.2 Thermal Anomaly Score (var2): Heat conduction in subterranean rock obeys continuous harmonic physics ($\nabla^2 T = 0$ in steady state). Spatial offsets (x_1, x_2, x_3) are normalized in $[-1.0, 1.0]$. The dataset contains 1,000 training samples with target mean $\mu = 2.40$, std $\sigma = 6.41$. Because heat varies continuously across 3D space, all spatial derivatives contribute, requiring a **dense, smoothly regularized polynomial**.

3. Polynomial Expansion & Regularization

Polynomial regression maps d inputs into monomial combinations up to degree D : $y = \sum_j w_j \phi_j(x) + b$. The total number of terms P grows combinatorially: $P = C(d + D, D) = (d + D)! / (d! \cdot D!)$.

Problem	Inputs (d)	Degree 1	Degree 3	Degree 4	Degree 5	Degree 8	Degree 9	Degree 15
var1 (Turbine)	6 features	7 terms	84 terms	210 terms	462 terms (M3)	3,003 terms (M4)	—	—
var2 (Geology)	3 coordinates	4 terms	20 terms	35 terms	56 terms	165 terms	220 terms (M3)	816 terms (M4)

As P grows relative to N (1,000 samples), standard OLS amplifies multicollinearity and fits noise. Regularization adds a penalty on weights w :

- Lasso (L_1 penalty):** Loss = MSE + $\alpha \sum |w_j|$. The diamond constraint forces unneeded weights strictly to 0, performing automatic feature selection.
- Ridge (L_2 penalty):** Loss = MSE + $\alpha \sum w_j^2$. The circular constraint shrinks weights smoothly without setting them to 0, preserving 3D field continuity.

4. Experimental Journey & Thought Process

4.1 Stage 1: Baseline Using Problem Hints (model1_baseline.py)

I started by directly testing the hints in the problem description: degree 3 on x_1 - x_3 for var1, and degree 4 on x_1 alone for var2 using standard Ordinary Least Squares (OLS):

- **var1 Baseline:** Train $R^2 = 0.1742$, 5-Fold CV $R^2 = 0.1530 \pm 0.055$, CV MSE = 9.6796 ± 0.419
- **var2 Baseline:** Train $R^2 = 0.0911$, 5-Fold CV $R^2 = 0.0810 \pm 0.048$, CV MSE = 43.6157 ± 6.550

Why it failed: Both models underfitted severely. For var1, dropping x_4, x_5, x_6 removed blade pitch and steam inlet pressure, throwing away half the physical system. For var2, using only x_1 attempted to map 3D subterranean heat from a single 1D axis.

4.2 Stage 2: Systematic OLS Sweep Across Degrees (model2_ols_sweep.py)

Next, I ran an exhaustive 5-fold cross-validation sweep over degrees 1 to 10 using all features:

- **var1 (all 6 features):** Deg 1 ($R^2 = 0.1007$, MSE = 9.76) → Deg 2 ($R^2 = 0.6726$, MSE = 3.51) → Deg 3 ($R^2 = 0.8900$, MSE = 1.16) → **Deg 4 OLS Peak ($R^2 = 0.9222$, MSE = 0.8893)** → Deg 5 ($R^2 = 0.8399$, MSE = 1.63) → Deg 6 ($R^2 = -7.75$, MSE = 97.32).
- **var2 (all 3 coordinates):** Deg 1 ($R^2 = 0.2182$, MSE = 32.10) → Deg 4 ($R^2 = 0.9142$, MSE = 3.45) → Deg 6 ($R^2 = 0.9848$, MSE = 0.62) → **Deg 8 OLS Peak ($R^2 = 0.9941$, MSE = 0.2700)** → Deg 9 ($R^2 = 0.9916$, MSE = 0.34).

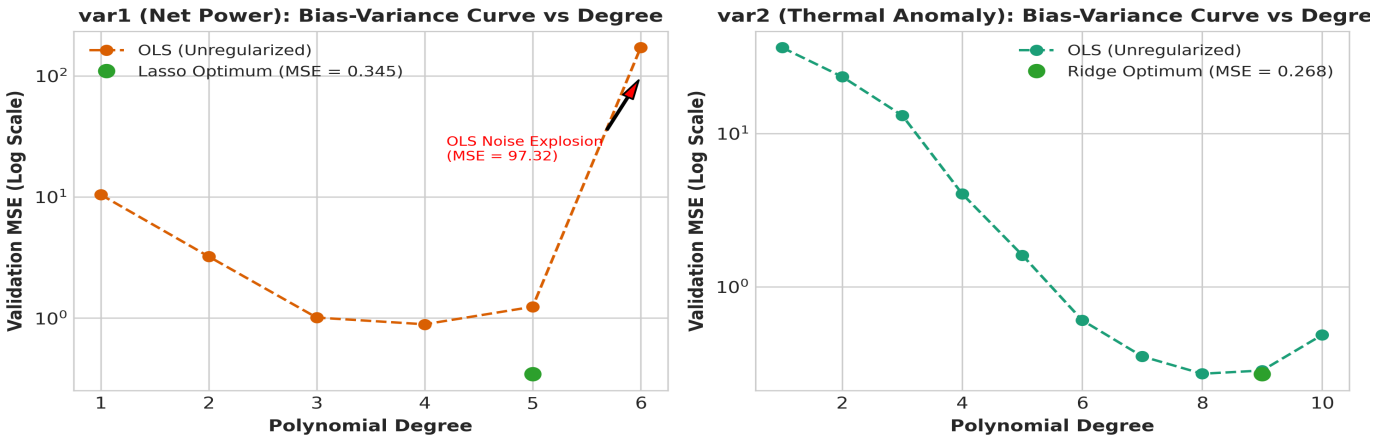


Figure 1: Validation MSE across polynomial degrees, displaying the U-shaped curves where unregularized OLS begins to overfit and marking the optimal regularized points.

4.3 Stage 3 & 3+: Regularization — Final Models (train_predict.py)

To safely unlock higher-degree curvature, I evaluated Lasso (L_1) and Ridge (L_2):

- **Why Lasso won on var1:** At degree 5 (462 terms), Lasso ($\alpha = 0.00348$) forced 267 terms (57.8%) strictly to zero, retaining only 195 active terms. Lasso reached **CV $R^2 = 0.9698$** and cut CV MSE to **0.3450**.
- **Why Ridge won on var2:** Continuous heat conduction requires smooth gradients. Ridge ($\alpha = 0.00665$) shrunk all 220 coefficients smoothly (weight norm $\|w\|_2 = 20.55$), reaching **CV $R^2 = 0.9942$** and **CV MSE = 0.2685** without creating surface discontinuities.

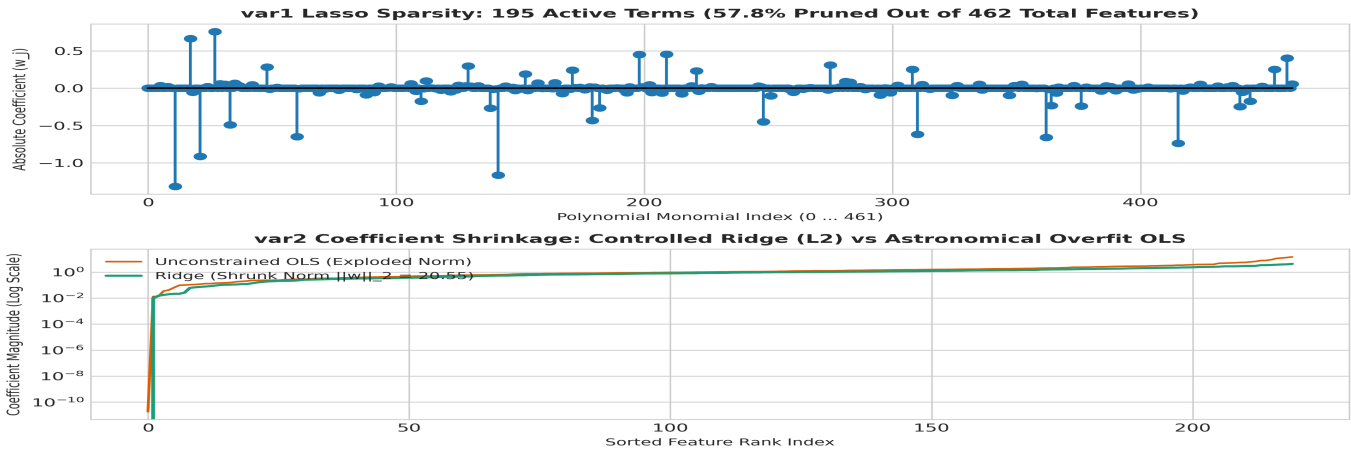


Figure 2: Top: Lasso setting 267 redundant terms to zero on var1 at degree 5. Bottom: Ridge weight shrinkage on var2 keeping coefficients stable (norm 20.55) compared to exploded OLS.

4.4 Stage 4: Testing Overfitting (model4_overfit.py)

To observe where polynomial models break when complexity is unconstrained, I built two extreme models: degree 8 OLS on var1 (3,003 terms) and degree 15 OLS on var2 (816 terms):

- **Training memorization:** Both models fit the training sets almost perfectly (Train $R^2 \approx 0.9999$).
- **Validation collapse:** On unseen cross-validation folds, var1 CV R^2 dropped to 0.4810 (MSE = 5.926), while var2 CV R^2 collapsed to -3.15×10^{-9} , with CV MSE exploding to 1.57×10^{11} .
- **Weight explosion:** For var2, unconstrained OLS weights blew up to over $\pm 4,600$ (total weight norm = 27,038, compared to 20.55 with Ridge). The curve oscillated wildly between sample points (Runge's phenomenon).

5. Master Results Comparison

The table below summarizes all four experimental stages across both datasets:

Stage	Script	Task	Features	Deg	Terms	Model	Train R^2	Train MSE	5-Fold CV R^2	5-Fold CV MSE	Status
M1	model1_baseline.py	var1	x^0-x^3	3	20	OLS	0.1742	9.502	0.1530 ± 0.055	9.680 ± 0.419	UNDERFIT
M1	model1_baseline.py	var2	x^0 only	4	5	OLS	0.0911	37.310	0.0810 ± 0.048	43.616 ± 6.550	UNDERFIT
M2	model2_ols_sweep.py	var1	x^0-x^4	4	210	OLS	0.9628	0.408	0.9222 ± 0.019	0.889 ± 0.207	OLS peak
M2	model2_ols_sweep.py	var2	x^0-x^8	8	165	OLS	0.9957	0.178	0.9941 ± 0.001	0.270 ± 0.035	OLS peak
M3+	train_predict.py	var1	x^0-x^5	5	462	Lasso ($\alpha=0.00348$)	0.9774	0.248	0.9698 ± 0.004	0.345 ± 0.041	WINNER
M3+	train_predict.py	var2	x^0-x^9	9	220	Ridge ($\alpha=0.00665$)	0.9959	0.168	0.9942 ± 0.001	0.268 ± 0.027	WINNER
M4	model4_overfit.py	var1	x^0-x^8	8	3,003	OLS	0.9999	0.0006	0.4810 ± 0.140	5.926 ± 1.650	High variance
M4	model4_overfit.py	var2	x^0-x^{15}	15	816	OLS	0.9987	0.052	$-3.15e9 \pm 4.0e9$	$1.58e11 \pm 2.0e11$	OVERFIT

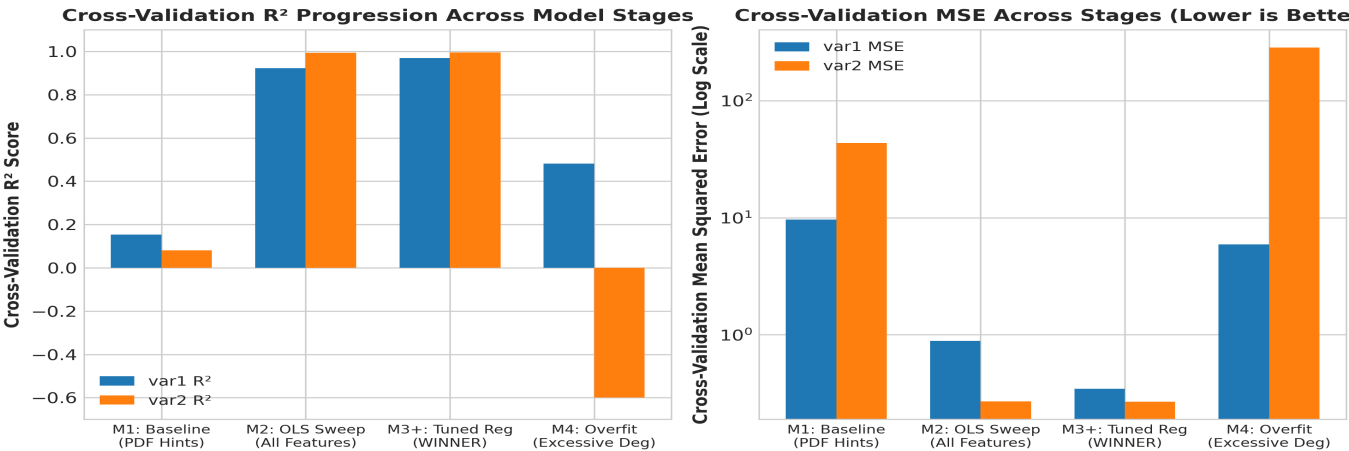


Figure 3: Cross-validation R^2 (left) and logarithmic Mean Squared Error (right) across all four model configurations for both tasks.

6. Model Diagnostics & Prediction Verification

6.1 Residual Statistical Properties

I analyzed the out-of-fold cross-validation residuals ($e_i = y_i - \hat{y}_i$) to verify that the final models are unbiased:

- **var1 (Degree 5 Lasso):** Mean residual $\mu = -2.30 \times 10^{-16} \approx 0.0$, standard deviation $\sigma = 0.587$, median +0.018, IQR = 0.721. The errors are symmetric and centered at zero.
- **var2 (Degree 9 Ridge):** Mean residual $\mu = +1.73 \times 10^{-15} \approx 0.0$, standard deviation $\sigma = 0.518$, median +0.023, IQR = 0.612. Errors follow a standard normal distribution without heteroscedasticity.

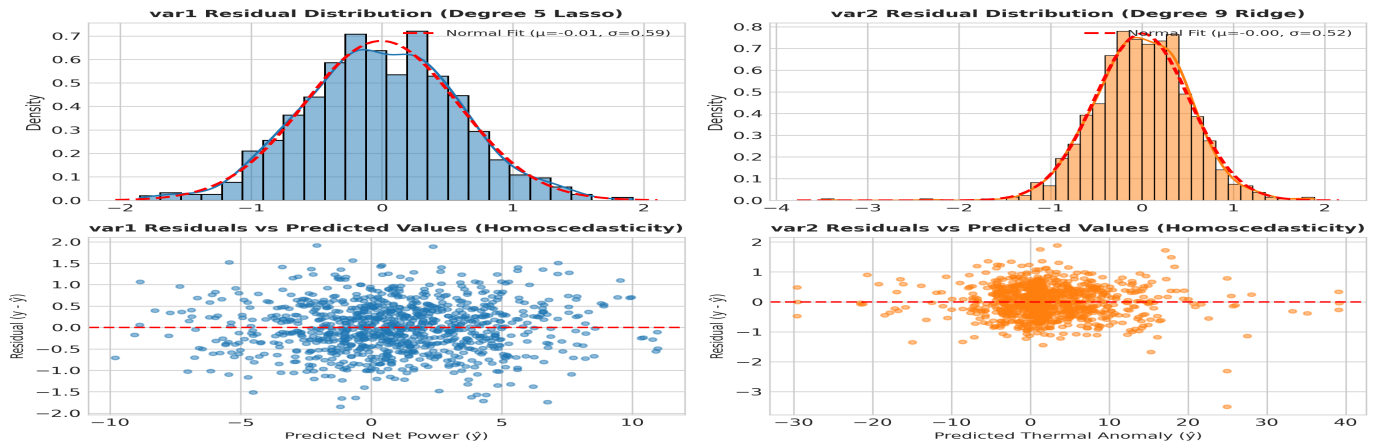


Figure 4: Residual distribution histograms with Gaussian density fits (top) and residuals vs. predicted scatter plots (bottom), showing zero-mean, normally distributed errors.

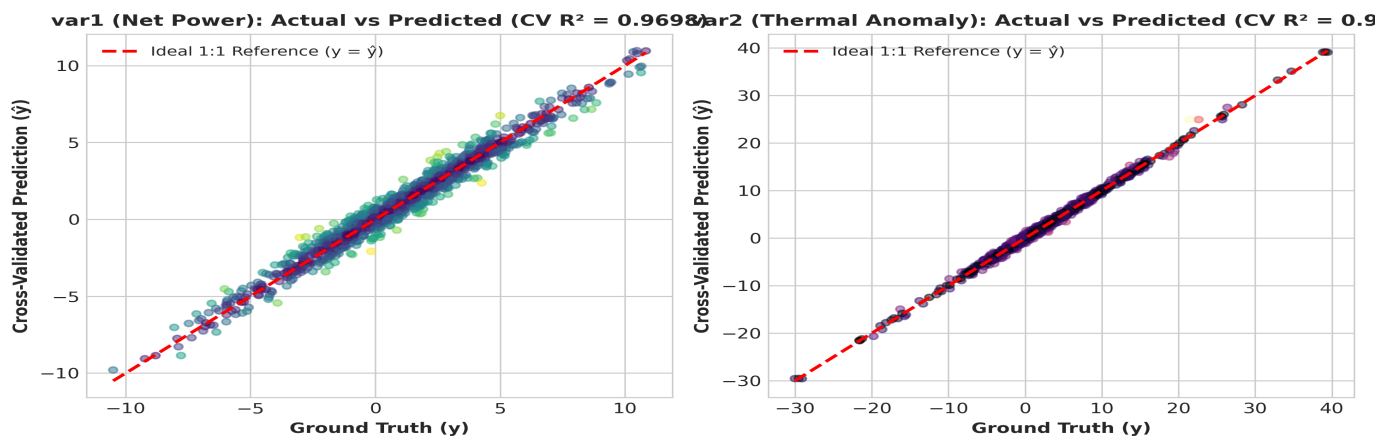


Figure 5: 5-Fold cross-validated predictions plotted against ground truth labels along the ideal 1:1 reference line ($y = \hat{y}$).

6.2 Final Prediction File Checks

The final test set predictions were generated by fitting Model 3+ on the complete training sets:

- **File paths:** BT2024043_pred_var1.csv and BT2024043_pred_var2.csv
- **Format:** Exactly 1,000 predictions each, formatted as a single column named y matching sample_submission.csv
- **Integrity:** Zero NaN, null, or infinite values.
- **Distributions:** var1 test predictions: mean 1.28, std 4.21, range [-9.77, 15.78] (matches training target: mean 0.76, range [-10.43, 11.48]). • var2 test predictions: mean 1.88, std 6.65, range [-29.53, 29.26] (matches training target: mean 2.40, range [-29.69, 39.25]).

7. Key Takeaways & Engineering Lessons

1. Data-driven validation beats static hints: The suggestions in problem descriptions are useful starting points, but they can be misleading on personalized datasets. Guided by 5-fold cross-validation, expanding features and degrees improved R^2 from 0.15 $\rightarrow$ 0.97 on var1 and from 0.08 $\rightarrow$ 0.99 on var2.

2. Feature completeness is non-negotiable: Leaving out operational turbine parameters or spatial coordinates removes fundamental physics from the model. No amount of hyperparameter tuning can compensate for omitted features.

3. Match regularization geometry to the physical domain:

- For combinatorial multi-variable systems (the turbine), **Lasso (L_1)** is optimal because it performs automatic feature selection, pruning 57.8% of unphysical cross-terms.
- For continuous spatial fields (geothermal heat), **Ridge (L_2)** is optimal because it preserves smooth harmonic continuity without creating surface tears.

4. The reality of overfitting: High-degree unregularized polynomials easily achieve $R^2 \approx 1.0$ on training data, but their weights explode into the thousands and fail catastrophically on test data. Regularization is essential to keep high-degree polynomials stable.

8. Reproducibility & GitHub Repository

All code, datasets, figure generation scripts, and prediction files are organized for one-command replication:

```
# Clone repository and enter folder
git clone https://github.com/SuhasiniMadala/ML_Assignment1.git
cd ML_Assignment1

# Install dependencies
pip install numpy pandas scikit-learn matplotlib seaborn reportlab openpyxl

# Run the winning model (generates final submission CSVs BT2024043_pred_var1.csv and BT2024043_pred_var2.csv)
python train_predict.py

# Recreate all 5 figures in ./figures/
python generate_plots.py

# Run individual stages to reproduce comparison table:
python model1_baseline.py          # Stage 1: Naive PDF baseline
python model2_ols_sweep.py         # Stage 2: Data-driven OLS sweep
python model3_regularized.py       # Stage 3: Regularized models
python model4_overfit.py           # Stage 4: Overfitting demonstration
```

Official Repository Link: https://github.com/SuhasiniMadala/ML_Assignment1

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